

# PRATYUSH DHUNGANA

Geospatial Data Scientist & GIS Automation Engineer | Forest Biometrics | Timberland Planning

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## PROFESSIONAL SUMMARY

Geospatial data scientist and GIS automation engineer with a forestry and natural-resources foundation and an M.S. in Forest Biometrics. Build the custom ArcPy toolboxes, automated geoprocessing pipelines, enterprise GIS workflows, and statistical models that turn LiDAR, remote sensing, and forest-inventory data into operational decisions — across forestry research at Mississippi State’s Forest Biometrics Lab and utility-scale industry. Also experienced in field-based forest resource assessment, forest management planning, timber valuation, and harvest scheduling, and in small area estimation for forest biomass and carbon.

## AREAS OF EXPERTISE

**Forest Management & Timber Economics:** Forest management planning · Silviculture · Stand dynamics · Growth & yield · Harvest scheduling & optimization (linear programming) · Timber valuation · DCF / NPV / LEV · 10-year management plans

**Biometrics, Estimation & Modeling:** Forest inventory & mensuration · Height–diameter modeling · Sampling design · Small area estimation · Aboveground biomass & carbon estimation · Statistical & machine-learning models · Model validation · Uncertainty assessment

**Geospatial & Remote Sensing:** GIS · High-density LiDAR · Satellite & aerial imagery · Google Earth Engine · Multi-temporal change analysis · Enterprise GIS (ArcGIS Portal, feature & REST services) · Cartography

**Programming, Automation & Data:** Python · ArcPy toolbox development · R (lidR, terra, tidyverse) · SQL · geoprocessing automation · FastAPI · R Shiny · JavaScript (React, D3) · PostGIS / SQLite / R-tree · GDAL · PDAL · Git

## PROFESSIONAL EXPERIENCE

### Associate Geospatial Analyst

Dec 2025 – Present

*The Spear Group, on behalf of Florida Power & Light · West Palm Beach, FL*

- Developed automated LiDAR-to-raster workflows using Python and ArcPy to quantify existing vegetation volume and projected removal volume across electric-utility corridors, supporting vegetation-management planning.
- Produced spatial distribution maps of vegetation volume and removal requirements, translating LiDAR-derived measurements into location-specific information for operational prioritization.
- Engineered a full-stack LiDAR data retrieval application using Python, R Shiny, SQLite, and R-tree spatial indexing to automate area-of-interest tile discovery, retrieval, and batch export — reducing multi-day data-staging workflows to minutes.
- Automated processing of 14,000+ high-resolution LiDAR tiles for raster generation, feature extraction, and large-scale utility vegetation analysis.
- Co-led a 38,616+ mile positional-accuracy assessment of electric-utility network data using LiDAR-derived ground truth, spatial statistics, and multi-ring buffer analysis across transmission, feeder, and lateral assets.
- Developed spatial analyses for vegetation-encroachment forecasting, drone placement, and road reachability by integrating LiDAR, remote sensing, spatial statistics, and machine learning.
- Built and documented 10 custom ArcPy geoprocessing tools, including end-to-end user tutorials and technical documentation.
- Designed geospatial schemas and published feature services through an ArcGIS Enterprise Portal environment; used ArcGIS REST services to access enterprise GIS resources and service-based data workflows.
- Supported enterprise application operations for GEVI (GE Visual Intelligence) — a platform for planning and scoping arborist and vegetation-management work — as assistant to the project manager: coordinated user roles and permissions, submitted and tracked vendor support tickets, attended recurring data-ingestion, user-feedback, and release-planning meetings, and contributed to release notes and user status updates.

## Graduate Teaching & Research Assistant

Jan 2024 – Dec 2025

Forest Biometrics Lab, Mississippi State University · Starkville, MS

- Conducted graduate research developing small area estimation methods for county-level aboveground forest biomass, combining FIA forest-inventory data with remote-sensing-derived auxiliary information — the statistical foundation of forest carbon-stock quantification.
- Developed and evaluated statistical and machine-learning models to estimate forest attributes across geographic domains, assessing performance with quantitative accuracy and uncertainty measures.
- Built reproducible Python and R workflows automating preprocessing, quality control, statistical analysis, and model evaluation for large forest-inventory and remote-sensing datasets.
- Co-instructed undergraduate laboratories in forest inventory, sampling design, stand dynamics, statistics, regression, and GIS.

## Research Assistant

Sep 2023 – Dec 2023

NAAMII · Lalitpur, Nepal

- Led a three-person team through ground-truth collection, geospatial data preparation, image labeling, and model development and evaluation for automated tree-species classification from high-resolution satellite imagery using YOLOv5.

## Forestry Consultant

Apr 2023 – Jun 2023

Independent · Nepal

- Conducted preliminary resource assessments with forest officials and local stakeholders, including participatory resource mapping, to identify forest resources, management priorities, and operational considerations across 30+ community forests.
- Conducted forest-boundary surveys, developed field sampling designs, and collected inventory data to quantify standing forest resources.
- Analyzed inventory and spatial information to support management recommendations and standing-stock revenue estimates using established timber prices.
- Prepared formal 10-year forest management plans and submitted reports to the Division Forest Office; plans were reviewed and approved for implementation as official management plans.

## GIS / Forestry Intern

Feb 2022 – Jun 2022

Forest Research and Training Center / Division Forest Office · Myagdi, Nepal

- Designed a geodatabase for 30+ community forests, integrating scanned maps, field surveys, digitized features, and aerial photography.
- Developed and compared machine-learning models for land-cover classification from remotely sensed data.
- Led an 8-person field team through a 20-day bat acoustic monitoring and spatial-data collection campaign, and ran a GIS workshop for 300+ high-school students on burned-area detection and mapping from satellite imagery.

## Remote Intern

Jan 2021 – Jul 2021

Friends of Nature (FON Nepal)

- Designed questionnaires, analyzed survey data, and prepared conservation reports on wildlife perception and conservation status.
- Organized conservation outreach for 150+ high-school students and developed educational materials for local communities.

## SELECTED PROJECTS

- **FIA Tree Height Codes: Crews vs. a Height Model** — independent analysis: trained height–diameter models on 11.8M FIA trees and evaluated ~200,000 crew-reconstructed heights on broken-top trees, diagnosing a spurious stump-height artifact by simulation. Public write-up and code.
- **Carbon Data Atlas** (2026) — built and deployed an open web atlas (React, D3) cataloging 16 satellite, model, inventory, and ground-based CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O datasets, with coverage maps and metadata export; [pratyush-dh.github.io/carbon-data-atlas](https://pratyush-dh.github.io/carbon-data-atlas).
- **County-Level Forest Biomass Estimation** — evaluated conventional and modern small area estimation approaches for aboveground forest biomass using FIA inventory data and remote-sensing-derived auxiliary variables; results presented at four national and regional research meetings.
- **Timberland Harvest Scheduling & Revenue Optimization** — evaluated a commercial forest tract under alternative management constraints and built a harvest schedule to maximize revenue, applying timber values, planning-period constraints, linear programming, NPV, and LEV to compare management strategies.

- **Python GIS Toolbox Development** — built ArcGIS Pro toolboxes integrating ArcPy, Spatial Analyst, and 3D Analyst to standardize utility GIS tasks, with documentation and user guides for team adoption.
- **Drone-Placement Suitability & Optimization** — integrated FAA, FEMA, environmental, and parcel data into a multi-criteria spatial model using cost-accumulation and spatial-overlap analysis.

## EDUCATION

### Master of Science — Forest Biometrics

Dec 2025

Mississippi State University · Starkville, MS · GPA 4.0

- Thesis: *Small Area Estimation Methods for County-Level Forest Biomass Estimation*.
- Relevant coursework: Advanced Forest Inventory · Forest Biometrics · Forest Resources Management · GIS Programming · Geodatabase Design · Statistical Computations in R · Machine Learning in Life Sciences.

### Bachelor of Science — Forestry

Oct 2022

Institute of Forestry, Tribhuvan University · Pokhara, Nepal · Aggregate 80.80%

- Relevant coursework: Forest Resources Economics · Forest Mensuration · Forest Statistics · Experimental Design · GIS & Remote Sensing.

## PRESENTATIONS

- County-Level Biomass Estimation Across Four FIA Units. *Partnership for Small Area Estimation Meeting*, 2025.
- Enhanced Biomass Estimation at Smaller Domains: Conventional vs. Modern Small Area Estimation Techniques. *Biannual Southern Silviculture Research Conference*, 2025.
- County-Level Aboveground Forest Biomass Estimation Using Remote-Sensing-Derived Auxiliary Information. *Graduate Student Research Symposium*, 2025.
- Machine Learning Approaches to Enhance Small Domain Estimates. *FIA Science Symposium*, 2024.

## LEADERSHIP

### Secretary / Coordinator

May 2019 – May 2022

Bat Friends Pokhara · Nepal

- Led an 11-member undergraduate team in bat research and monitoring; coordinated conservation and outreach programs across five districts.

## TECHNICAL PROFICIENCIES

**Analysis:** Python · R · SQL · pandas · NumPy · statistical modeling · machine learning

**Forestry / GIS:** ArcGIS Pro · QGIS · Google Earth Engine · ArcPy · lidR · terra · LiDAR · remote sensing · spatial analysis

**Data & Infrastructure:** PostgreSQL/PostGIS · SQLite / R-tree · geodatabases · GDAL · PDAL · FastAPI · R Shiny · Git/GitHub

## TRAINING & CERTIFICATIONS

- Species Distribution Modeling with Remote Sensing — NASA ARSET webinar.
- Satellite Observations and Tools for Fire Risk, Detection, and Analysis — NASA ARSET webinar.
- Fundamentals of GIS — University of California, Davis (Coursera).
- Python Basics — Michigan State University (Coursera).
- Spatial Data Science: The New Frontier in Analytics — Esri MOOC.
- Going Places with Spatial Analysis — Esri MOOC.
- Wildlife Research Techniques Training, 2020 (19 days) — Friends of Nature (FON), Nepal.

## HONORS

- Graduate Student Award, Xi Sigma Pi — Alpha Theta Chapter, 2025.
- Tribhuvan University Full Tuition Waiver, 2018–2022.